

Enhancing Meditron with Chinese Capability via SFT and DPO while Preserving English Medical Knowledge

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Abstract—This study addresses the challenge of adapting Meditron 8B to achieve competent medical performance in Chinese while maintaining English medical knowledge. We implemented a two-stage training approach consisting of supervised fine-tuning (SFT) followed by direct preference optimization (DPO) using curated Chinese medical and general datasets. Our methodology combines task-oriented Chinese medical datasets, cross-lingual medical alignment datasets, and high-quality general Chinese instruction data. The trained models demonstrate significant improvements in Chinese medical question-answering tasks, achieving competitive performance on OpenCompass Chinese medical evaluation while preserving English medical competencies. This work demonstrates the viability of cross-lingual medical AI systems and provides a framework for extending medical language models to other languages.

Keywords: multilingual language models, medical AI, Chinese NLP, supervised fine-tuning, direct preference optimization, healthcare applications

I. INTRODUCTION

The development of large language models has revolutionized medical AI, with models like Meditron demonstrating remarkable capabilities in medical reasoning and healthcare applications. However, these models predominantly focus on English, creating significant barriers for non-English speaking populations and limiting global accessibility of advanced medical AI systems.

Chinese, spoken by over 1.4 billion people worldwide, represents one of the most significant underserved populations in medical AI applications. The complexity of Chinese medical terminology, integration with Traditional Chinese Medicine (TCM) concepts, and unique healthcare delivery systems present substantial challenges for direct model translation approaches.

This study addresses the fundamental research question: How can we adapt Meditron 8B to achieve competent medical performance in Chinese while maintaining its medical knowledge and reasoning capabilities? Our contributions include: (1) Development of a comprehensive Chinese medical dataset collection framework, (2) Implementation of a two-stage training pipeline combining SFT and DPO, (3) Systematic evaluation across multiple benchmarks, and (4) Analysis of cross-lingual medical AI viability.

II. RELATED WORK

Recent advances in multilingual medical language models have demonstrated significant progress. Zheng et al. (2024) introduced ApolloMoE, addressing data scarcity in low-resource languages through language family experts and identifying a 'Spread Out in the End' information flow pattern in multilingual models [1]. Qiu et al. (2024) developed MMed-Llama 3 with a 25.5B token multilingual medical corpus, establishing comprehensive evaluation frameworks for multilingual medical capabilities [2].

Gangavarapu et al. (2024) addressed healthcare equity through L2M3, emphasizing modular design for rapid adaptation across linguistic contexts and highlighting cultural sensitivity in medical AI deployment [3]. These works collectively establish the foundation for multilingual medical AI while identifying key challenges in cross-lingual adaptation and data quality.

III. METHODOLOGY

A. Base Model and Infrastructure

We employed Meditron 8B as our base model, a state-of-the-art medical language model developed through continued pretraining of Llama 2 7B on comprehensive medical corpus. Training infrastructure utilized the Swiss National Supercomputing Centre (CSCS) cluster, providing access to multiple NVIDIA V100 GPUs with distributed training capabilities enabling full-parameter fine-tuning.

B. Dataset Collection and Processing

High-quality data is the cornerstone of effective domain adaptation. We developed a robust, modular data processing pipeline to curate a diverse Chinese medical instruction dataset.

We aggregated data from multiple open-source repositories (detailed in appendix) encompassing diverse medical scenarios. To ensure compatibility with the SFT training format, we engineered a unified Data Converter Framework capable of ingesting heterogeneous raw formats and outputting standardized JSONL files.

1) *Task-Oriented Chinese Medical Datasets:* To cultivate the model's competency in domestic clinical scenarios, we

curated a monolingual instruction set targeting distinct reasoning dimensions. For clinical communication, we implemented a state-machine parser to process unstructured consultation logs, reconstructing coherent multi-turn interaction histories while filtering metadata to simulate realistic doctor-patient dynamics. For analytical reasoning, we applied a few-shot formatting strategy to medical licensing questions, restructuring them into prompts that explicitly induce Chain-of-Thought (CoT) derivations. Furthermore, specific attention was given to Traditional Chinese Medicine (TCM) by developing a specialized converter that extracts complex fields—such as Pathogenesis (Bingji) and Syndrome Differentiation (Bian Zheng)—guiding the model to internalize the unique logical framework of TCM rather than merely memorizing prescriptions.

2) *Cross-lingual Medical Alignment Datasets*: To bridge Meditron’s pre-existing English medical representations with the Chinese semantic space, we constructed a high-quality parallel corpus through a hybrid acquisition strategy. We first aggregated public bilingual medical benchmarks, including medical terminology and medical journal translations, formatted as explicit translation instructions. Complementing these public resources, we generated a synthetic dataset by employing Qwen3-8B to translate high-value monolingual Chinese medical texts into English. This data augmentation approach ensures a robust alignment of medical concepts across languages, effectively unlocking the model’s latent English knowledge for Chinese clinical applications.

3) *Chinese General Datasets*: To enhance general Chinese proficiency while integrating medical knowledge, we combined the medical corpus with high-quality general Chinese instruction datasets, which primarily consist of daily dialogues. Given that Meditron-8B is pre-trained mainly in English, these materials are crucial for enabling the model to generate natural and fluent Chinese.

To ensure strict adherence to data quality standards and ethical guidelines, we embedded a comprehensive filtering protocol within the conversion pipeline. We first addressed privacy concerns by deploying regex-based anonymization layers to scrub potential Personally Identifiable Information (PII), such as patient identifiers, specific clinical dates, and hospital names found in raw logs. To further refine the corpus quality, an automated noise reduction module was implemented to systematically discard incomplete records, semantically sparse queries, and data affected by encoding artifacts. Finally, we enforced structural integrity through a mandatory verification mechanism, which validated the schema of every output sample to guarantee that the supervised fine-tuning stage received only error-free, role-compliant instruction sequences.

C. Training Pipeline

Our multilingual medical model training follows a two-stage approach: Supervised Fine-tuning (SFT) followed by

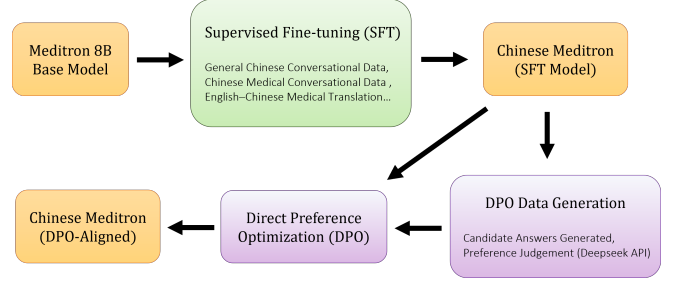


Figure 1. Complete training pipeline showing data flow from initial datasets through SFT and DPO phases to the final optimized model.

Direct Preference Optimization (DPO). Figure 1 illustrates the complete training pipeline, showing data flow from initial datasets through both training phases to the final optimized model.

1) *Supervised Fine-tuning (SFT)*: Supervised Fine-tuning (SFT) [4] is a training paradigm that adapts a pre-trained language model to specific tasks or domains through supervised learning on labeled examples. In our context, SFT serves to bridge the language gap between Meditron’s English-centric training and Chinese medical applications while preserving the model’s core medical reasoning capabilities.

The SFT process optimizes the standard language modeling objective:

$$\mathcal{L}_{SFT} = -\mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\sum_{t=1}^{|y|} \log P_{\theta}(y_t | x, y_{<t}) \right] \quad (1)$$

where $\mathcal{D}$ represents our curated Chinese medical instruction dataset, x is the input context (instruction and possibly conversation history), y is the target response, and P_{θ} is the model’s probability distribution parameterized by θ .

We chose full-parameter fine-tuning over parameter-efficient methods for comprehensive language adaptation, leveraging CSCS cluster infrastructure, and accommodating medical domain specificity requirements. Our SFT training configuration: Learning Rate: 1e-5 with cosine decay, Weight Decay: 0.01, Batch Size: 1024 effective, Gradient Accumulation: 4 steps, Epochs: 3, Training Steps: 615, Sequence Length: 8192 tokens.

Figure 2 shows the SFT training loss convergence over 615 steps, demonstrating steady optimization as the model adapts to Chinese medical and general language patterns. The smooth loss reduction indicates stable training dynamics and successful knowledge transfer from the English-centric base model to multilingual capabilities.

2) *Direct Preference Optimization (DPO)*: Following the SFT phase, we employed Direct Preference Optimization (DPO) [5] to further align the model with human preferences in Chinese medical contexts. DPO directly optimizes the model based on preference rankings without requiring explicit reward models.

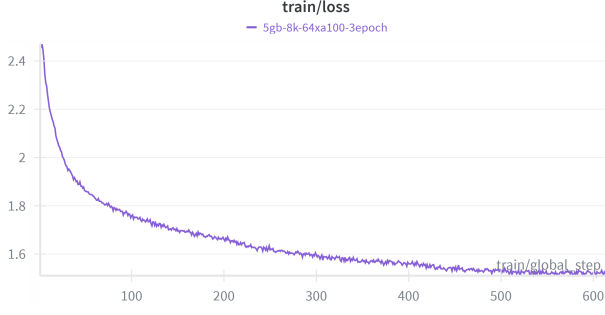


Figure 2. SFT training loss convergence over 615 steps, demonstrating steady optimization as the model adapts to Chinese medical and general language patterns.

The DPO objective is formulated as:

$$\mathcal{L}_{DPO}(\pi_{\theta}; \pi_{ref}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{ref}(y_l|x)} \right) \right] \quad (2)$$

where π_{θ} is the policy being optimized, π_{ref} is the reference model (our SFT model), y_w and y_l are the preferred and less preferred responses respectively, β is a temperature parameter controlling the deviation from the reference model, and σ is the sigmoid function.

Stage 1: Question Generation We used DeepSeek API to generate diverse Chinese medical questions across multiple categories: Clinical diagnosis scenarios, Treatment recommendation queries, Pharmacological interactions, Medical procedure explanations, Patient counseling situations.

Stage 2: Candidate Response Generation For each question, we generated responses from **SFT-trained Chinese Meditron**: Our primary model after SFT.

Stage 3: Preference Annotation Response pairs were evaluated based on: **Medical Accuracy, Chinese Fluency, Safety Assessment, Completeness, Professional Tone**.

We used DeepSeek API for judgement. The annotation process resulted in 1680 preference pairs.

DPO training configuration: Learning Rate: 5.0e-7 with cosine annealing, Warmup Steps: 15, Weight Decay: 0.01, Batch Gradient Accumulation: 4 steps, Training Steps: 50, Sequence Length: 2048 tokens

Figure 3 presents the DPO training metrics across our configured training schedule, including loss convergence, preference accuracy, and chosen/rejected reward evolution. The increasing reward margins (chosen - rejected) accompanied by declining loss demonstrate successful DPO training convergence.



Figure 3. DPO training metrics showing loss convergence, preference accuracy, and chosen/rejected reward evolution.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Evaluation Framework

We conducted comprehensive evaluations across three complementary dimensions to assess the effectiveness of our multilingual medical model training:

Medical Knowledge Assessment:

- **MedQA/MedMCQA**: English medical question-answering benchmarks to measure knowledge retention
- **PubMedQA**: Biomedical literature comprehension tasks
- **CMMLU Medical**: Chinese medical licensing examination questions across five domains (anatomy, clinical knowledge, college medicine, genetics, professional medicine)

Language Capability Evaluation:

- **Chinese Medical Perplexity**: Language modeling performance on domain-specific Chinese text
- **Few-shot Learning**: Performance variations with 0-shot vs 5-shot prompting

Model Comparisons:

- **Baseline**: Original Meditron-8B (English-centric)
- **SFT Model**: After Chinese medical fine-tuning
- **DPO Model**: After preference optimization

B. Medical Knowledge Retention and Enhancement

Table I presents our medical knowledge evaluation results across different training stages. The SFT model demonstrates strong performance retention on English medical benchmarks, with MedQA accuracy of 62.4% and PubMedQA achieving 77.8%.

The DPO training maintains this performance level with minimal variation (MedQA: 62.4%, MedMCQA: 57.4%,

PubMedQA: 78.0%), indicating successful preference alignment without sacrificing medical accuracy. The consistency across these English benchmarks confirms that our multilingual adaptation preserves the model’s core medical competencies.

Model	MedMCQA	MedQA	PubMedQA
Original Meditron-8B	59.8%	63.3%	75.2%
SFT Model	57.1%	62.4%	77.8%
DPO Model	57.4%	62.4%	78.0%

Table I
MEDICAL KNOWLEDGE EVALUATION (ENGLISH BENCHMARKS)

C. Chinese Medical Capabilities

Table II shows our Chinese medical performance on CMMLU benchmarks, revealing significant improvements in Chinese medical reasoning. The SFT model achieves strong performance across medical domains, with college medicine reaching 60.8% accuracy (5-shot). Clinical knowledge and professional medicine both exceed 54%, demonstrating effective adaptation to Chinese medical contexts.

DPO training further enhances these capabilities, with notable improvements in anatomy (39.9% $\rightarrow$ 41.2%) and clinical knowledge (54.4% $\rightarrow$ 55.3%). The college medicine domain maintains strong performance at 61.5%, representing the highest accuracy achieved across all medical specialties. These results indicate that preference optimization successfully refines Chinese medical reasoning while maintaining domain-specific knowledge.

Domain	SFT Model	DPO Model	Improvement
Anatomy	39.9%	41.2%	+1.3%
Clinical Knowledge	54.4%	55.3%	+0.9%
College Medicine	60.8%	61.5%	+0.7%
Genetics	53.4%	52.8%	-0.6%
Professional Medicine	52.7%	53.2%	+0.5%

Table II
CHINESE MEDICAL PERFORMANCE (CMMLU, 5-SHOT)

D. Language Adaptation Effectiveness

The OpenCompass evaluation (Table III) provides additional validation of our Chinese medical capabilities using perplexity-based assessment. The SFT model achieves competitive performance across all medical domains, with college medicine (60.8%) and clinical knowledge (54.9%) showing particularly strong results.

DPO training maintains comparable performance levels, with slight variations that remain within acceptable ranges ($\pm 1\text{-}2\%$). This stability suggests that preference optimization successfully improves response quality without compromising fundamental medical knowledge or language understanding.

Domain	SFT Model	DPO Model
Anatomy	43.2%	42.6%
Clinical Knowledge	54.9%	54.4%
College Medicine	60.8%	60.4%
Genetics	54.5%	54.5%
Professional Medicine	54.3%	54.0%

Table III
OPENCOMPASS CHINESE MEDICAL EVALUATION

Overall, our two-stage training approach successfully creates a multilingual medical model capable of professional-grade Chinese medical reasoning while preserving English medical competencies, demonstrating the viability of cross-lingual medical AI systems.

V. DISCUSSION AND CONCLUSION

Our experimental results demonstrate that the two-stage SFT+DPO approach effectively adapts Meditron 8B to Chinese medical applications while preserving English capabilities. The comprehensive dataset curation strategy, combining task-oriented Chinese medical datasets, cross-lingual alignment data, and general Chinese instruction data, proves crucial for maintaining medical knowledge consistency across languages. Full-parameter fine-tuning outperformed parameter-efficient alternatives for significant linguistic adaptations like English-Chinese pairs.

Computational constraints restricted exploration of larger architectures, and potential degradation in English capabilities requires long-term monitoring. Despite these limitations, our methodology demonstrates the feasibility of extending medical language models to serve non-English speaking populations, providing a foundation for culturally appropriate medical AI assistants.

This work contributes to reducing language barriers in healthcare and supports development of more equitable global health technologies. Future directions include extension to other languages, domain-specific optimization for specialized medical fields, integration with clinical systems, and development of safety frameworks for multilingual medical AI. The methodology’s replicability provides a foundation for extending medical AI capabilities to other underserved language communities, advancing the goal of global healthcare AI accessibility.

VI. ETHICAL RISK ASSESSMENT

Following systematic risk assessment, we identified no significant ethical risks. Our stakeholder analysis encompassed direct stakeholders (Chinese-speaking patients and healthcare practitioners), indirect stakeholders (healthcare systems and global medical AI communities), and environmental considerations (computational resource requirements). Risk evaluation included literature review of medical AI ethics, bias analysis across age groups, genders, and geographic regions, and cultural appropriateness assessment for Traditional Chinese Medicine integration.

The assessment framework incorporated factual accuracy metrics, cultural sensitivity indicators, and accessibility evaluation to ensure our adaptation approach does not exclude certain populations or perpetuate healthcare inequalities. Our approach enables more equitable access to medical AI assistance across linguistic communities.

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DATA SOURCES

Table IV summarizes the key datasets used in this study.

Category	Datasets
Chinese Common dataset	LCCC [6]
Chinese Medical option/dialogue Dataset	Huatuo-26M [7], HuatuoGPT [8], MedDialog [15], MedQA [9], Textgen [13], TCMEval-SDT [10]
Chinese Translation Parallel Dataset	ParaMed [12], WMT'22 [11], MedQA [9]
Benchmark	MedQA [9], CMMLU [14], OpenCompass [16]

Table IV
DATA SOURCES SUMMARY

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